

Comparing Pre-attentive Visual Variables in a Target Identification Task for Glanceable Visualizations

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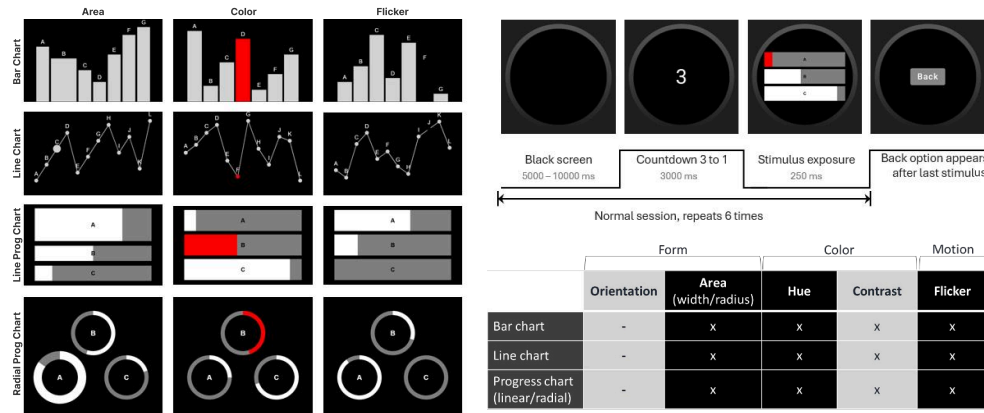


Figure 1: (Left) Examples of stimuli showing combinations of the four charts with the three pre-attentive visual variables. (Right, top) Procedure of an individual study session. (Right, bottom) Table showing the three selected visual variables (in black) and the two discarded ones (in gray), the x sign means the visual variable can be applied to the chart, and - it cannot.

ABSTRACT

In this paper, we evaluated the use of three pre-attentive visual variables, namely, area, color hue, and flicker, to highlight elements on four charts: bar chart, line chart, linear progress chart, and radial progress chart. We exposed 48 participants—using short time lapses of 250 ms—to a set of stimuli showing one of the four charts with the three visual variables. Then we examined the accuracy of their answers in identifying the highlighted target element under sedentary and walking conditions. Our results show that all three visual variables can be perceived with a medium to high accuracy rate, though performance varies depending on the mobility condition, the visual variable, and even the chart type. Among the visual variables, color and flicker yield the best target identification accuracy rates, while the area proves to be the least effective. The findings from our first investigation contribute to the design of efficient smartwatch micro visualizations that support glanceable and in-motion data reading. Supplemental material is available [here](#).

Index Terms: Pre-attentive visual variables, micro visualization, mobile visualizations, smartwatch.

1 INTRODUCTION

The use of pre-attentive visual variables, such as color hue or value, position, and motion, in data visualization is important, as these features facilitate the rapid detection of salient information—in less than 250 ms. These features are processed unconsciously in spatial memory [5, 15], which was proven to improve the cognitive process in visualization comprehension [4]. Grioui et al. [9] found that

reading small progress charts on a watch face while walking—in particular through obstacles—increases the mental load of a task and can be more physically demanding at high speeds. Consequently, the need for rapid detection of information and guidance becomes important to reduce the lookup time when viewers are under additional mental or physical load from concurrent tasks.

Historically, pre-attentive visual variables focused on desktop and static settings [1], including a few works that evaluated their use for providing quick feedback in the case of peripheral glanceable visualizations [16]. In the latter study, the authors found that among six primitive visual tasks, the horizontal position was the best, and color luminance was the worst. However, previous work still lacks insights on how visual variables can be transferred to small screens such as smartwatches and, in particular, when read in motion. Despite their small screens, smartwatches show various types of data—on average 5 and up to 17 complications on a watch face [13]. The data can be represented as small-scale charts—also called micro visualizations [11]. The displayed information is designed to be easily glanceable (i.e., viewed in a few milliseconds) and is usually consumed on the go, such as while exercising. However, with such a small and packed display, identifying a single element (e.g., abnormal heart rhythm on a time series visualization) can be a time-intensive process if not highlighted properly. For this reason, we believe that micro visualizations can benefit from the use of pre-attentive visual variables to enable quick data assessment and to reduce interruption from a primary activity.

The efficiency of pre-attentiveness varies between visual variables [19] and depends on the chart's representation and structure. Hence, we conducted an empirical evaluation to assess which of the selected visual variables are more efficient and accurate for smartwatches' small display area and mobility, while performing a target identification task. Our goal was not to compare the four visualizations because they have different representations and serve different purposes; rather, we assess the efficiency of these pre-attentive visual variables for each visualization.

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2 STUDY

In our study, pre-attentive visual variables are used as design components i. e., exploring their capability—“their visual attention characteristics”—as part of the visualization design [1]. We conducted a mixed (factorial) design study to assess the effectiveness of three visual variables when used with micro visualizations in two mobility conditions. The chart-type variable was evaluated in a between-subjects design, and the pre-attentive visual variables in a within-subjects design using a walking and a baseline sedentary condition. The study was conducted using a Google Pixel Watch 2 with a round screen of 41 mm.

2.1 Task and Stimulus Design

The goal of the study task was to simulate a micro-visualization reading task on a smartwatch. Therefore, it was important to focus on the most common chart types that a regular smartwatch would show. Then, we chose the visual variables, which can be applied to the chosen chart types (Figure 1, right bottom).

2.1.1 Chart Type Selection

People from different ages and educational backgrounds use smartwatches [18]. To accommodate this diversity and facilitate personal data interpretation, we chose visualizations that the general public would consider to be common and widely recognized. Progress charts, either linear or radial, are common visualizations on smartwatches. They were used in prior works [3, 9, 20] to depict the progress toward a goal, specifically for fitness data. One popular example are the Apple Watch activity rings. Additionally, bar and line charts are typical visualizations used to show data on smartwatches. Bar charts are used to compare different data points such as daily sleep data [12]. Whereas line charts are usually used for time-series data, for instance, heart-rate measurements over time [17]. Therefore, in our study, we focus on these four chart types to investigate the reading of pre-attentive visual variables on smartwatches.

2.1.2 Three Visual Variables

Based on Ware [19] the following visual features—color, orientation, size, contrast, and motion or blinking—have the most substantial effect. We started with these five visual variables, which were subsequently filtered based on their suitability to be used with our four charts (Figure 1, right bottom). We eliminated the orientation visual variable because applying it to the circles in the line chart would not have an effect. Moreover, tilting one bar of the bar chart or a progress bar would distort the chart structure and make it hard to read. Our final selection of pre-attentive visual variables consists of area (width for bar and progress charts and radius for the line chart circles), color hue, and flicker.

2.1.3 Visualization Stimulus Design

All stimuli are displayed in the center of the smartwatch screen. For a maximum exploration of the screen space, we opted for a juxtaposed layout—which is a common representation of complications on smartwatch faces [3, 9]—to represent each individual line and radial progress chart as an independent component. Besides the alphabetic labels displayed above and on the charts, no other description was included (see Figure 1, left).

We generated six unique datasets and created six stimuli for each chart type. The number of data points per dataset varies based on the typical usage scenarios of chart designs on smartwatches. The bar chart was designed to show 7 data points referring to the 7 weekdays, the line charts showed 12 data points referring to half a day in hours or the 12 months, and for the progress charts, we displayed them in a group of three, which is common for fitness data displayed on watch faces [3, 9, 20]. Each data point in the stimulus was associated with an alphabetical letter [A - G] for bar charts, [A - L] for line charts, and [A, B, C] for the progress charts. For both

line and bar charts, the data is ordered by position along a common scale, making the identification of the first and last elements (bar or point) easy to read compared to the rest of the elements. For these two charts, we therefore deliberately chose not to highlight the first and last elements. Each stimulus was displayed on the smartwatch for 250 ms and after each exposure participants were asked to speak out loud the letter corresponding to the highlighted target element. Participants' answers were noted in an Excel sheet. Each participant performed a total of 36 trials (1 chart type $\times$ 6 stimuli $\times$ 3 visual variables $\times$ 2 conditions).

We ensured that our dataset did not contain data with progress proportions of 0% and 100%. The data values ranged between 10 and 95 for the bar and line charts and 10% and 90% for the progress charts. When choosing the target element, we ensured that they were equally distributed both on the left side or top (elements [B - C] for the bar chart, [B - E] for the line chart, and A for both the progress charts), in the middle (elements D for the bar chart, F and G for the line chart, and B for the progress charts), and on the right side or bottom (elements E and F for the bar chart, [H - K] for the line chart, and C for the progress charts). This allowed us to test different positions of the target element and avoid learning effects because positions were chosen arbitrarily.

We opted for a black background and neutral shades for the elements (bar and circles). We used a light gray shade (#D3D3D3) for bars and line chart circles. We used a darker gray shade to represent the empty part of a progress chart (#808080) and white (FFFFFF) to represent the filled proportion. We chose white instead of the light gray shade because it offers a better contrast with the dark gray shade. We encoded the three pre-attentive visual variables for the four visualizations as follows:

Color hue encoding: We used red (#FF0000) to highlight a target element because it is a bright color, and in general, it is much easier to find, in particular when used for visual search tasks [14].

Size encoding: We doubled the size (width and radius) of the highlighted target element compared to the baseline size.

Flicker encoding: We created the flickering effect by alternating between two images. The first image showed a baseline stimulus (without any highlight), while the second showed the target element in black (the background color to simulate a fading effect). We created the flicker by switching the images every 83 ms (within the total 250 ms) in a way that the first image (without highlighting) is displayed twice and the second image (with highlighting) once.

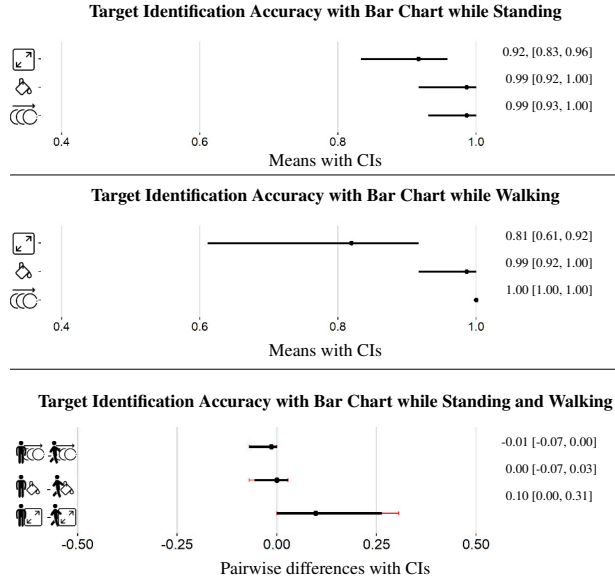
2.2 Participants

In total, we recruited 48 participants (23 F and 25 M). Twenty-nine participants were aged between 18 and 24, 16 were aged between 25 and 34, one participant belonging to the 35 to 44 age range, and two participants in the 45 to 54 age range. Exactly half of the participants had some form of vision deficiency, which was either corrected by contacts or glasses. Most participants' (31) highest degree was a high school diploma, 10 had a bachelor's degree, 6 participants had a master's degree, and one participant had a doctorate (PhD). Out of 12 participants for each group, 12 reported that they are familiar to very familiar with bar charts, 9 with line charts, and 8 with the two progress charts.

2.3 Study Procedure

After reading and signing the privacy and consent forms, participants were introduced to the study task. One out of the four possible chart types was randomly assigned to participants. Next, we presented further details related to their assigned chart-type task. Then, participants were asked to wear the smartwatch on their preferred wrist and perform a training round showing two stimuli for each visual variable. After confirming that participants understood the task and got familiar with using the smartwatch, the study began.

Table 1: Means of target identification accuracy while standing (Row 1) , while walking (Row 2) , and pairwise differences (Row 3) for the **bar chart**. area, color, flicker



Depending on the study condition order, participants either started walking at a normal pace (not too slow or too fast) or stood at a specific spot in the study room. When a stimulus appeared, the participants called out the letter corresponding to the highlighted target element in the visualization. In case of doubt about the spelled letter, a naming scheme (i.e., the NATO phonetic alphabet) was used before noting the answer. During each session, participants were shown six stimuli alternated with a black screen. When the smartwatch screen turned black, participants were not looking at the watch. They were notified with a vibration and a beep sound three seconds before the next stimulus was displayed (Figure 1, right top). At the end, we asked participants to fill out a post-study questionnaire to report on their experience with the chosen visualization and visual variables. Then, participants provided some demographic information. On average, participants completed the study within 30 minutes and were compensated monetarily (minimum wage). The university ethics board approved the study.

3 RESULTS AND DISCUSSION

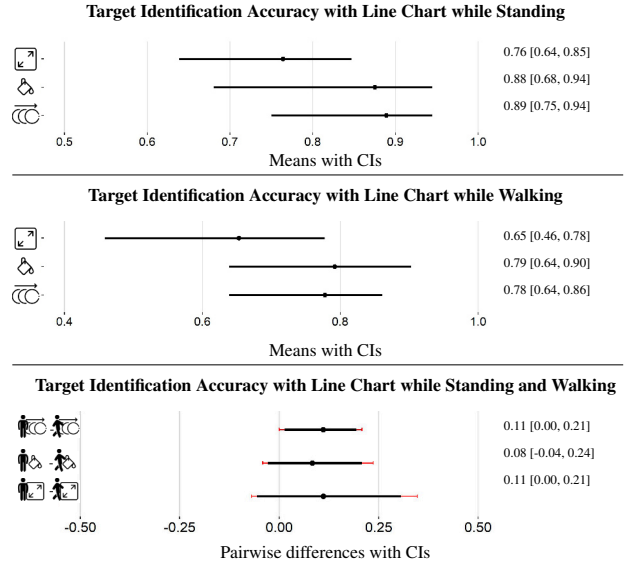
In this section, we report on our results, including the target identification accuracy rate of the highlighted target element, in addition to participants' confidence and aesthetics rating for the three visual variables in the two mobility conditions.

3.1 Target Identification Accuracy

For the target identification accuracy, we report sample means for 1,728 trials in total: 432 trials (12 participants $\times$ 6 stimuli $\times$ 3 visual variables $\times$ 2 conditions) for each visualization using 95% confidence interval (CI) estimation [2, 6, 7, 8] using BCa bootstrapping (10,000 bootstrap iterations). In addition, we report the differences between means to compare the three visual variables. We adjusted the CIs of mean differences for multiple comparisons using Bonferroni corrections [10] highlighted as red error bars. We interpret a non-overlap of the CIs with 0 as evidence of a difference.

The Impact of the Three Visual Variables. On average, participants were able to identify the highlighted target accurately for the three visual variables: while **standing** area [line: 76% - linear & radial prog: 98.6%], color [line: 88% - linear prog: 100%], and flicker [line: 89% - linear & radial prog: 100%] and while **walk-**

Table 2: Means of target identification accuracy while standing (Row 1) , while walking (Row 2) , and pairwise differences (Row 3) for the **line chart**. area, color, flicker



ing area [line: 65% - linear prog: 100%], color [line: 79% - linear & radial prog: 100%], and flicker [line: 78% - linear prog: 98.6%]. Our results show evidence of differences and trends that the answers with the area were less accurate than the color and the flicker visual variables for the bar chart (with a mean difference of 7% while standing and $\approx$ 18% while walking) and for line chart (with a mean difference of $\approx$ 13% while standing and 14% only with the color while walking; Table 1, 2, 3, and 4).

Even though our goal was not to compare the four visualizations, our results show clear evidence that identifying the target item with the line chart was more challenging. This implies that the size of the circles might have an influence on efficiency of the pre-attentiveness of the visual variables on small displays. Moreover, according to participants, the short display time affected participants' ability to read the associated letters, especially when the target element was on the right side. In such cases, participants had to recall the position of the highlighted item (e.g., 10th point or 3rd starting from the right side) to then provide the letter (e.g., j) accordingly. A further investigation of the results showed that with the line chart, 47% of the errors happened with targets on the right side of the graph (H - K) followed by 32% on the left side, and 21% in the middle. This shows that central items were less challenging than the ones on both sides, because they might be aligned with the initial gaze direction—the center of the display. However, this confirms our assumption that guessing the letter for the items on the right side based on the alphabetical order required more mental load and influenced, to some extent, the correctness of the provided answers.

The Impact of the Mobility Conditions. We compared the target identification accuracy for each visual variable between the two mobility conditions (see Row 3 in Table 1, 2, 3, and 4). The walking condition did not significantly impact participants' ability to identify the highlighted target element. However, on the one hand, the results show trends that participants' answers were slightly less accurate for the area while walking (by 10% on average) with the bar chart and with the flicker (by 11% on average) with the line chart, and (by 1% and 3%) with the linear and radial progress chart. Participants' feedback was in accordance with these findings, one mentioned "walking made both area and especially motion noticeably more difficult, since I can't hold the watch completely stable."

Table 3: Means of target identification accuracy while standing (Row 1), while walking (Row 2), and pairwise differences (Row 3) for the **linear progress chart** area, color, flicker

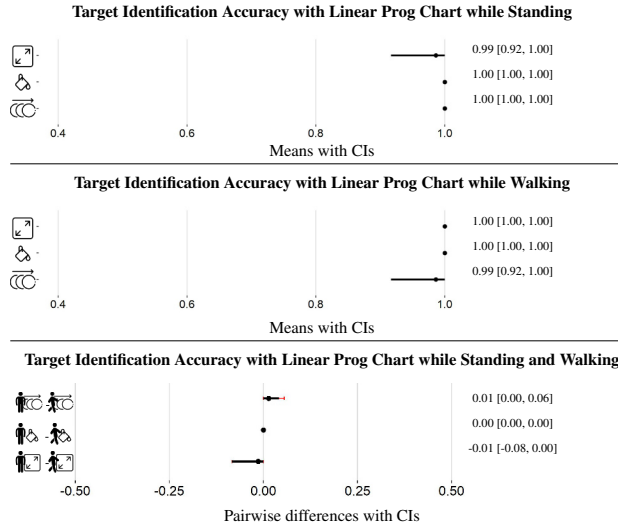
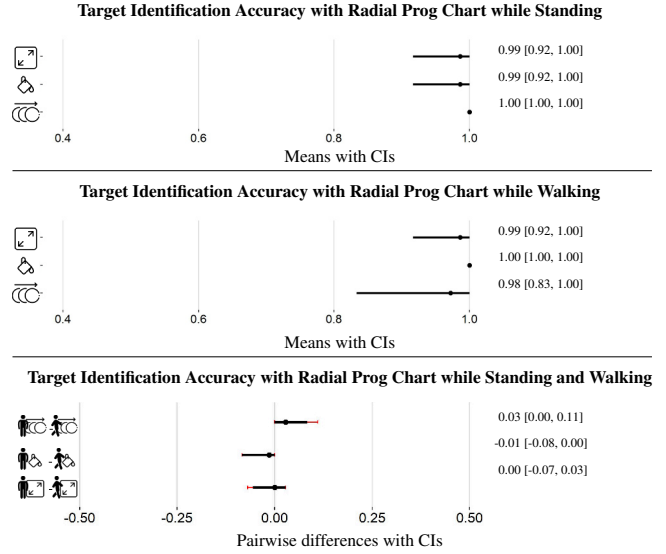


Figure 2: Confidence level rating for the three visual variables.

Table 4: Means of target identification accuracy while standing (Row 1), while walking (Row 2), and pairwise differences (Row 3) for the **radial progress chart** area, color, flicker



On the other hand, an opposite effect with a minimal difference of 1% was noticed for the bar chart with the flicker, the radial progress chart with the color, and the linear progress chart with the area.

3.2 Preference Ratings

We asked participants to rate their preference for the visual variables based on aesthetics and confidence level with their answers.

Confidence with Answers. In this section we report on the total count for *confident* and *very confident* votes. In general, the color visual variable was always ranked the highest (bar chart: 9 $\times$, line chart: 6 $\times$, line and radial progress charts: 12 $\times$). In contrast, area received more negative ratings, in particular with the line (2 $\times$) and bar (4 $\times$) charts, followed by flicker with the line chart (4 $\times$) (see Figure 2). According to participants, the efficiency of the area visual variable was lower than the color and flicker. Some of them found that the difference in radius between the highlighted and non-highlighted circles in the line chart was insufficient. Others mentioned that with the bar chart, they had to go for a sequential search “it was harder to see because I had to look at almost every bar.” The problem was accentuated with visualizations showing rectilinear shapes such as the bar chart and linear progress charts, in which the bar length is used to encode a value or a progress. Using the bar width to encode additional information created a conjunction of variables, which led to a more mentally demanding search task for them. One participant wrote “It is hard to accomplish the task when the differences between the bars are height and width at the same time.” In addition, some of them highlighted the possible risk of misinterpretation when the bar height and width are simultaneously used for encoding information.

Preference for Aesthetics. Participants found color more aesthetically pleasing and helpful with the performed task for all the visualizations (see supplemental material). According to participants, the red color was catchy, and it created a more substantial contrast effect compared to the two other visual variables. They also found it more comfortable to the eye. With the radial progress chart, participants’ ratings were split between color (with 7/12 votes) and area (with 5/12 votes). One participant mentioned: “color was too stunning [...] and area was moderate.” Similarly to the area visual variable, flicker received 5 votes: one for the linear progress chart and four for the bar chart.

Summary: We summarize take-away messages from the study.

- **color:** Among the four visualizations, color is ultimately the most effective, the most preferred overall, and the least affected by motion.
- **flicker:** To reduce the visual discomfort, the flicker motion should be used carefully: 1) avoid applying it to items that occupy an important portion of the display ($\geq 1/3$ of the screen) and 2) avoid very fast flicker frequencies.
- **flicker:** The effect is likely to be unnoticeable with very small elements (like circles in a line chart).
- **area:** To increase the efficiency of the pre-attentive effect on small displays, the area of the highlighted target element should be significantly different from the distractors.
- **area:** We recommend avoiding the use of area (i. e., width) with rectilinear shapes because it might be confusing.

4 CONCLUSION

While color was the most effective, we consider that the flicker and area can be good alternatives with monochrome displays and when used properly. However, future works are required to address certain limitations, such as the scope of the tested pre-attentive visual variables, and their inherent variations (e. g., different color luminance and flicker frequencies). Additionally, it is important to assess to what extent the obtained results can be generalized to denser visualizations, and when shown at different sizes, layouts (e. g., concentric radial charts), and when performing other reading tasks (e. g., value comparison). Finally, a more suitable selection task, that does not impose additional cognitive burden, should be set.

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REFERENCES

- [1] L. Barrera-Leon, F. Corno, and L. De Russis. How the preattentive process is exploited in practical information visualization design: A review. *International Journal of Human-Computer Interaction*, 39(4):707–720, 2023. doi: 10.1080/10447318.2022.2049137 1, 2
- [2] L. Besançon and P. Dragicevic. The continued prevalence of dichotomous inferences at CHI. In *Extended Abstracts of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 1–11. ACM, New York, 2019. doi: 10.1145/3290607.3310432 3
- [3] T. Blascheck, L. Besançon, A. Bezerianos, B. Lee, A. Islam, T. He, and P. Isenberg. Studies of part-to-whole glanceable visualizations on smartwatch faces. In *IEEE Pacific Visualization Symposium (PacificVis)*, pp. 187–196. IEEE, 2023. doi: 10.1109/PacificVis56936.2023.00028 2
- [4] C. H. Chih and D. S. Parker. The persuasive phase of visualization. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, p. 884–892. ACM, New York, 2008. doi: 10.1145/1401890.1401996 1
- [5] W. S. Cleveland and R. McGill. Graphical perception: Theory, experimentation, and application to the development of graphical methods. *Journal of the American Statistical Association*, 79(387):531–554, 1984. doi: 10.2307/2288400 1
- [6] A. Cockburn, P. Dragicevic, L. Besançon, and C. Gutwin. Threats of a replication crisis in empirical computer science. *Communications of the ACM*, 63(8):70–79, 2020. doi: 10.1145/3360311 3
- [7] G. Cumming. *Understanding the new statistics: effect sizes, confidence intervals, and meta-analysis*. Routledge, 2013. 3
- [8] P. Dragicevic. Fair statistical communication in HCI. In J. Robertson and M. Kaptein, eds., *Modern Statistical Methods for HCI*, chap. 13, pp. 291–330. Springer, Berlin, Heidelberg, 2016. 3
- [9] F. Grioui, T. Blascheck, L. Yao, and P. Isenberg. Micro visualizations on a smartwatch: Assessing reading performance while walking. In *IEEE Visualization and Visual Analytics*, pp. 46–50. IEEE, 2024. doi: 10.1109/VIS55277.2024.00017 1, 2
- [10] J. Higgins. *Introduction to modern nonparametric statistics*. Thomson Learning, –, 2004. 3
- [11] P. Isenberg. *Micro visualizations: Design and analysis of visualizations for small display spaces*. Habilitation thesis, Université Paris-Saclay, Gif-sur-Yvette, France, 2021. 1
- [12] A. Islam, R. Aravind, T. Blascheck, A. Bezerianos, and P. Isenberg. Preferences and effectiveness of sleep data visualizations for smartwatches and fitness bands. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pp. 27:1–27:17. ACM, New York, 2022. doi: 10.1145/3491102.3501921 2
- [13] A. Islam, A. Bezerianos, B. Lee, T. Blascheck, and P. Isenberg. Visualizing information on watch faces: A survey with smartwatch users. In *IEEE Visualization and Visual Analytics*, pp. 156–160. IEEE, 2020. doi: 10.1109/VIS47514.2020.00038 1
- [14] D. Lindsey, A. Brown, E. Reijnen, A. Rich, Y. Kuzmova, and J. Wolfe. Color channels, not color appearance or color categories, guide visual search for desaturated color targets. *Psychological Science*, 21(9):1208–1214, 2010. doi: 10.1177/0956797610379861 2
- [15] J. Mackinlay. Automating the design of graphical presentations of relational information. *ACM Trans. Graph.*, 5(2):110–141, 1986. doi: 10.1145/22949.22950 1
- [16] J. Moreira, D. Mendes, and D. Gonçalves. Incidental visualizations: Pre-attentive primitive visual tasks. In *Proceedings of the International Conference on Advanced Visual Interfaces*, pp. 1–9. ACM, New York, 2020. doi: 10.1145/3399715.3399841 1
- [17] A. Neshati, F. Alallah, B. Rey, Y. Sakamoto, M. Serrano, and P. Irani. SF-LG: Space-filling line graphs for visualizing interrelated time-series data on smartwatches. In *Proceedings of the International Conference on Mobile Human-Computer Interaction*, pp. 1–13. ACM, New York, 2021. doi: 10.1145/3447526.3472040 2
- [18] P. Pawar and R. Jambhale. Smartwatch statistics by vendors, revenue and number of users. <https://electroiq.com/stats/smartwatch-statistics>. Accessed on 01.05.2025. 2
- [19] C. Ware. *Information Visualization: Perception for Design*. Morgan Kaufmann Publishers Inc., San Francisco, 3 ed., 2012. 1, 2
- [20] X. Zhouxuan, L. Yu, and P. Fabiola. A study on activity visualization for smart watches, 2024. doi: abs/2407.02012 2